

Networking Questions - Summary Answers

Q: Discuss the OSI (Open Systems Interconnection) model and explain the function of each layer. How does the layered architecture help in designing and understanding network communication protocols?

Answer Summary:

- 7 layers: Physical, Data Link, Network, Transport, Session, Presentation, Application.
- Physical: cables/signals; Data Link: error-free link (MAC); Network: IP/routing; Transport: TCP/UDP; Session: manages sessions; Presentation: formats/encrypts; Application: user services like HTTP.
- Benefits: easier troubleshooting, modularity, interoperability.

Q: Given a scenario where a user is unable to access a website, analyze how each layer of the OSI model could be used to troubleshoot the issue. Which layer would you check first and why?

Answer Summary:

- Start from Application if software-related, or Network for DNS/IP issues.
- Check each layer step-by-step to isolate the root cause.

Q: Define network transmission media and explain the differences between guided (wired) and unguided (wireless) media, giving examples of each.

Answer Summary:

- Guided (wired): Twisted Pair, Coaxial, Fiber - stable and secure.
- Unguided (wireless): Wi-Fi, Bluetooth - flexible but prone to interference.

Q: A company is setting up a network in a new building. The IT manager must choose between fiber-optic cables and twisted-pair cables. Analyze the factors that should influence this decision and justify which medium would be more suitable for a large enterprise with high data demands.

Answer Summary:

- Fiber-optic: high speed, long distance, no interference.
- Twisted pair: cheaper but slower.
- Fiber is better for large enterprises and future growth.

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Q: Imagine you are part of a team tasked with setting up a small network in a temporary outdoor event. You are responsible for selecting and justifying a transmission medium. How would you communicate your choice to non-technical team members to ensure understanding and team alignment?

Answer Summary:

- Choose Wi-Fi for fast setup.
- Explain with visuals/analogies.
- Mention pros (mobility) and cons (signal issues).
- Suggest wired backup.

Q: Explain the purpose of IEEE 802.11 in wireless LANs. What are the basic differences between IEEE 802.11 and Bluetooth in terms of range, application, and data rate?

Answer Summary:

- Wi-Fi: up to 100m, used for internet, high speed.
- Bluetooth: up to 10m, for device pairing, low speed.

Q: A user is experiencing frequent collisions in a wireless network. Analyze how the CSMA/CA mechanism works to reduce collisions and explain how control frames like RTS and CTS help solve issues such as the hidden station problem.

Answer Summary:

- CSMA/CA: listens before sending, avoids collisions.
- RTS/CTS reserve channel and prevent hidden node issues.

Q: You are helping set up a wireless network for a public event. How would you ensure optimal performance while explaining potential interference and connectivity limitations to event organizers with non-technical backgrounds?

Answer Summary:

- Explain interference in simple terms.

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- Use dual-band routers and good placement.
- Use analogy: Wi-Fi like sound in a crowded room.

Q: Describe the main fields found in an IEEE 802.11 wireless LAN frame. What is the purpose of each field?

Answer Summary:

- Frame Control: type; Duration: time; Addresses: source/destination;
Sequence Control: frame order; Frame Body: data; FCS: error check.

Q: Analyze a situation where a wireless LAN frame experiences transmission errors. Which parts of the frame are essential for detecting and managing these errors, and how does the system respond?

Answer Summary:

- FCS detects errors via CRC.
- If error found, frame is dropped and may be resent.

Q: Imagine you are training new employees on how wireless LAN communication ensures reliability despite interference and collisions. How would you explain the role of the frame structure in a simple, non-technical way?

Answer Summary:

- Compare frame to a package: labels = addresses, sticker = type, code = FCS.
- If damaged, resend the package. Helps reliable communication.